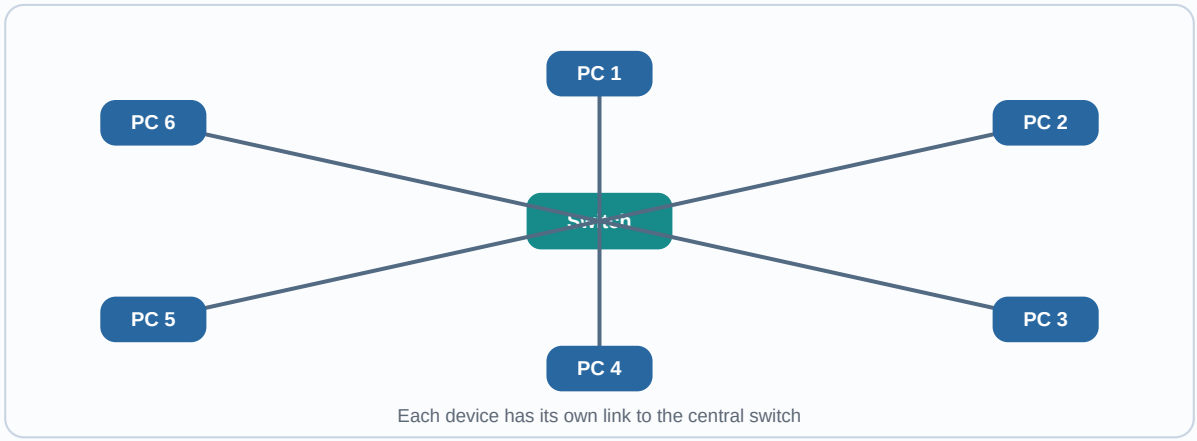
TASK 1 — Draw and Explain Network Topologies

A network topology describes how computers, servers, switches and other devices are connected. The three diagrams below show common physical or logical layouts. Use them as a guide when drawing the diagrams on paper or in a drawing tool.

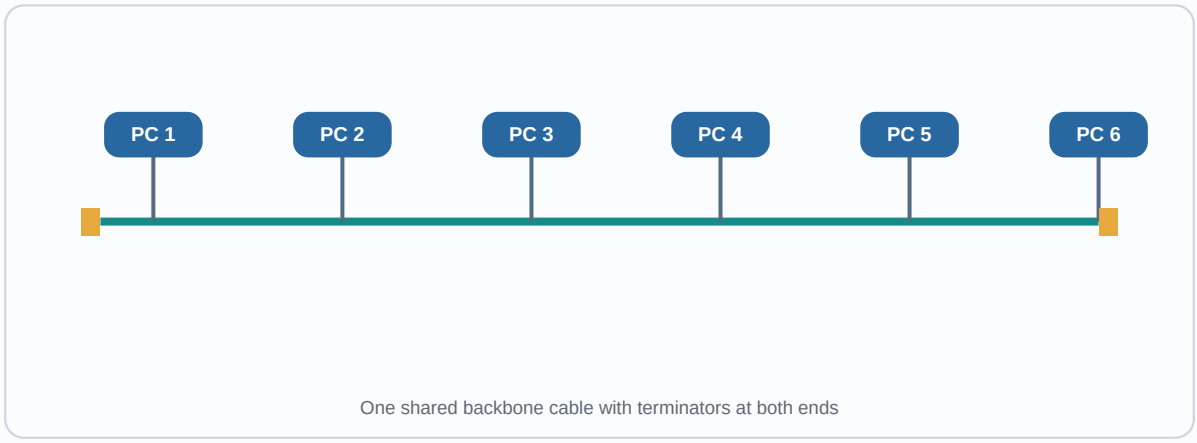
A. Star topology



Each device has its own link to the central switch

Real-world example: Flipkart warehouse office network. Workstations can connect to a central switch, which forwards traffic to the correct device. A faulty cable usually affects only one workstation, making faults easier to isolate.

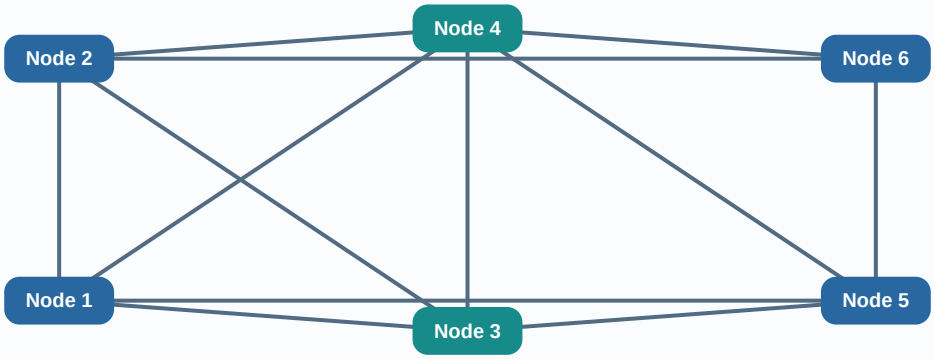
B. Bus topology



One shared backbone cable with terminators at both ends

Real-world example: a small, legacy shared-cable lab network. Multiple devices share one backbone cable. It uses less cabling, but a backbone fault can interrupt communication for many devices; modern Ethernet LANs generally prefer switches instead.

C. Mesh topology



Multiple links provide alternate paths between devices

Real-world example: resilient links between data-centre routers supporting WhatsApp-style messaging. Multiple paths can keep traffic moving if one link fails. The exact architecture of a commercial service is more complex than this simplified example.

Key point: the app name is an illustrative use case, not a claim that the service uses only one topology.

TASK 2 — Compare Four Network Topologies

The table lists at least four advantages and four disadvantages for each topology. The “Best Use Case” column includes familiar apps or services as examples of the type of system that may benefit; these are illustrations, not verified descriptions of their internal network designs.

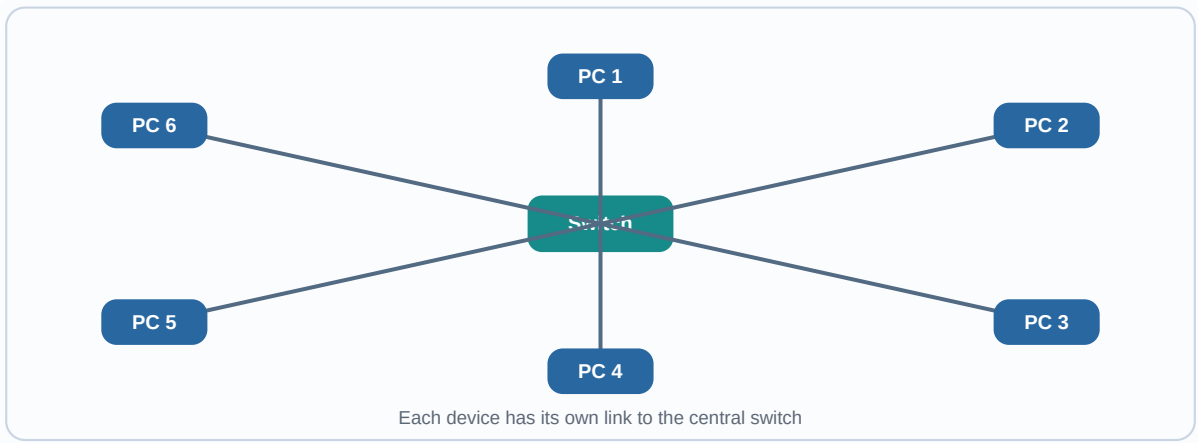
Topology	Pros (at least four)	Cons (at least four)	Best Use Case / Example
Star	• Easy to install and manage • Simple to add devices • One cable failure usually affects one device • Faults are easy to locate • Good performance with a switch	• Central switch is a single point of failure • Requires more cable than bus • Switch adds cost • Switch capacity can become a bottleneck • Needs power for active equipment	Office, school, shop or warehouse LAN. Example: computers used for Zomato order management connect to a central switch.
Bus	• Simple design • Uses less cable for a small setup • Low initial cost in legacy coaxial systems • No central switch required • Can be useful for learning basic shared-medium networking	• Backbone failure can stop the whole network • Performance drops as traffic increases • Faults can be difficult to isolate • Cable length and device count are limited • Collisions can occur in shared Ethernet designs	Small legacy or teaching networks. Example: a basic networking simulation demonstrating how shared access affects Paytm-style office traffic.
Ring	• Predictable data path • Each node can have a fair turn to transmit • Can avoid collisions in token-based designs • Orderly traffic flow • Dual-ring designs can offer resilience	• A single break may disrupt a simple ring • Adding/removing nodes can interrupt service • Troubleshooting may require checking the loop • Delay can rise as nodes are added • Less flexible than switched star LANs	Some industrial or metro/telecom ring systems. Example: a resilient communications ring carrying services used by Instagram offices or content infrastructure.
Mesh	• Multiple routes between nodes • High fault tolerance • Can reroute around a failed link • Supports reliable paths for critical traffic • Scales well when designed in layers	• Expensive to install • Requires many links and ports • Configuration is more complex • Troubleshooting can be complicated • Full mesh becomes impractical as node count grows	Backbone, data centre or wireless mesh. Example: resilient paths between systems supporting Instagram media delivery or messaging services.

How to choose: compare the budget, number of devices, expected traffic, tolerance for failure, ease of maintenance, and future expansion. The most reliable option is not always the cheapest or simplest.

TASK 3 — Design a Multiplayer Mobile Game Network

Scenario: You are designing a new multiplayer mobile game similar to PUBG or Free Fire. Players need quick updates about movement, actions, scores and match events. A short delay can affect the fairness and feel of gameplay.

Recommended design: a hierarchical star at each server location, with mesh-like redundant links between core routers and data centres.



Each device has its own link to the central switch

Why this design fits

1. **Speed:** Each game server and network device can have a direct high-speed link to a switch. This reduces unnecessary hops inside the local server network and supports low-latency communication.

2. **Reliability:** Redundant uplinks and alternate paths between core routers can keep a region connected if one link fails. Game servers can also be deployed across more than one availability zone or region.

3. **Cost:** A full mesh between every server would require many links and be expensive. A star inside each rack or site, combined with selective redundant backbone links, balances performance with cost.

4. **Scalability:** New servers can be added to switch ports or new racks. Matchmaking can assign players to a suitable regional server to reduce latency, while monitoring helps operators identify congestion.

Important design note

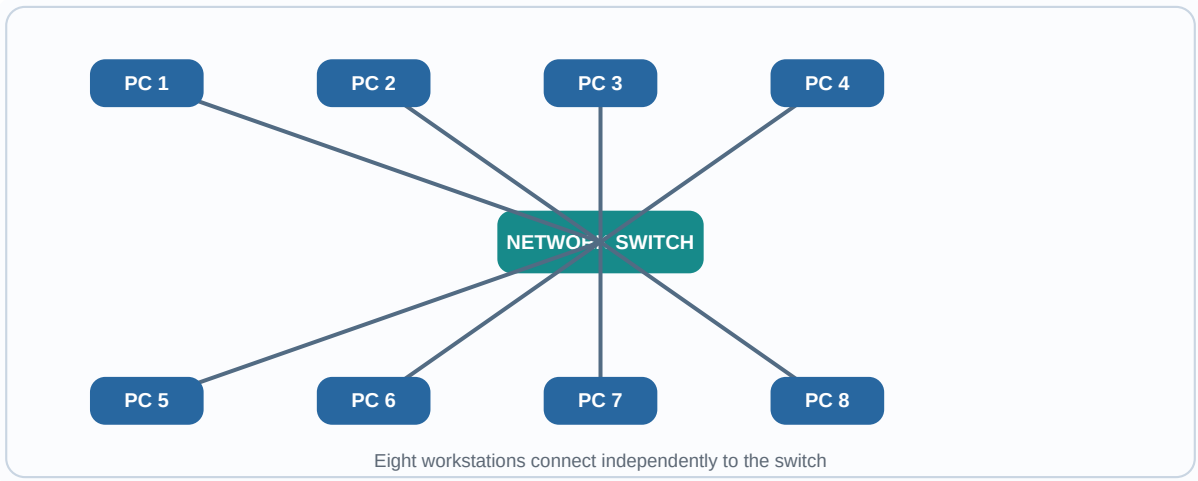
The game normally uses a client-server model: the phone sends player inputs to an authoritative game server, and the server validates actions and distributes state updates. Topology describes the connections between network devices; it does not by itself guarantee low ping. Server placement, routing, bandwidth, congestion control and anti-cheat design also matter.

Conclusion: choose a **star topology within each server rack** and **redundant mesh-like connections between core network devices**. This hybrid approach provides fast local connections, resilience and more manageable costs.

TASK 4 — Network for an Eight-Computer Warehouse

Scenario: A small Flipkart-style warehouse needs to connect eight computers used for order processing, stock updates, packing labels and inventory checks. The network should be affordable, easy to troubleshoot and simple to expand.

Recommendation: use a star topology with an Ethernet switch.



Eight workstations connect independently to the switch

Reasoning (2–3 lines)

Each of the eight computers connects to a central switch using its own Ethernet cable. If one cable or computer fails, the other workstations can usually keep working, and the faulty connection is easy to identify.

A 16-port switch would leave spare ports for future devices such as a printer, barcode scanner, access point or additional workstation. This gives the warehouse room to grow without redesigning the whole network.

Suggested setup checklist

- One 16-port Gigabit Ethernet switch
- Eight computers with Ethernet ports
- Eight Cat5e or Cat6 patch cables (plus spare cable)
- Router/firewall for internet access, if required
- Network printer or label printer, if required
- UPS for the switch/router where brief power interruptions are common
- IP addressing plan and a record of cable/port labels

Basic troubleshooting

If one computer cannot connect, check its cable, link lights, network adapter and switch port first. If all computers lose connectivity, check switch power, the router/uplink and any shared network equipment. Keep a simple port map so each desk can be traced to a switch port.

Security tip: use strong Wi-Fi security if wireless access is added, keep the router and computers updated, and restrict inventory-system access to authorised staff accounts.

EXTRA PAGE — How Topology Affects Everyday Services

Apps such as Instagram, Zomato, Paytm, WhatsApp and Flipkart depend on many layers of networking. Their complete infrastructure may include user devices, Wi-Fi or mobile networks, internet service providers, routers, load balancers, data-centre switches, servers and content-delivery networks. It is usually more accurate to discuss which topology suits a component than to assign one topology to an entire app.

Example 1: A shop or office

A switched star is a practical choice for desks, point-of-sale terminals and inventory computers. Each device has an independent cable to a switch, so moves and repairs are straightforward. A larger office can connect multiple access switches to distribution and core switches.

Example 2: A high-availability data centre

A data centre may use leaf-and-spine switching, which provides multiple paths between server racks and supports predictable bandwidth. This is not simply a textbook full mesh; it is a structured design that balances path diversity, cost and scale.

Example 3: A remote or difficult-to-wire area

A wireless mesh can extend coverage by allowing compatible access points or nodes to relay traffic. It may be useful in some campuses, warehouses or temporary sites, but wireless hops can reduce throughput and performance depends on interference and node placement.

Quick decision guide

Need	Often suitable	Why
Easy management for a small LAN	Star	Central switch and independent device links simplify changes and fault-finding.
Low-cost legacy shared medium	Bus	Simple cable layout, but a backbone fault has broad impact.
Ordered access on a loop	Ring	Can provide predictable transmission; resilience depends on ring design.
Alternate paths for critical links	Mesh / redundant paths	Traffic may reroute after a link failure, with added cost and complexity.

Remember: topology is one part of network design. Bandwidth, latency, redundancy, security, equipment quality and ongoing maintenance are equally important.

FINAL PAGE — Glossary, Summary and Conclusion

Glossary

Term	Meaning
Node	A device connected to a network, such as a computer, printer, switch or server.
Switch	A device that connects devices in a local area network and forwards Ethernet frames to the appropriate port.
Router	A device that forwards packets between different IP networks, such as a local network and the internet.
Backbone cable	The main cable used by a bus topology to carry shared network traffic.
Redundancy	Extra links or equipment that can help a network continue working after a failure.
Latency	The time taken for data to travel from a source to a destination; often measured in milliseconds.
Bandwidth	The amount of data a connection can carry per unit of time.
Single point of failure	A component whose failure can stop a larger part of the system from working.

Assignment summary

• **Task 1:** Star, bus and mesh diagrams were labelled and paired with realistic illustrative use cases.
• **Task 2:** Star, bus, ring and mesh were compared with at least four pros and four cons each.
• **Task 3:** A hybrid design was recommended for a multiplayer game: star inside server racks and redundant links between core devices.
• **Task 4:** A switched star was recommended for eight warehouse computers because it is manageable, easy to troubleshoot and expandable.

Overall conclusion

No single topology is best for every situation. Star is a strong default for small offices and warehouses because it is simple to maintain. Bus is mainly useful as a legacy or learning example, ring can suit specialised systems, and mesh or redundant paths improve resilience where availability is important. Good design matches the network to its purpose, budget and expected growth.

Submission reminder: add your name, class, roll number and date on the cover page. If your teacher requires hand-drawn diagrams, redraw the diagrams neatly on paper and attach them with the printed assignment.